Daeun Kim

dekim@casys.kaist.ac.kr | kimdaeun00.github.io

Research Interests

- Efficient generative AI system
- Model Quantization and Sparse computation
- GPU scheduling and resource management
- HW/SW Co-design

Education

KAIST, Ph.D. in School of computing — Mar. 2025 – date
- Advisor: Prof. Jongse Park

KAIST, M.S. in School of computing — Mar. 2023 – Feb. 2025
- Advisor: Prof. Jongse Park
- Thesis: Exploiting MX Quantization for Accelerating Diffusion Transformer-based Image Generative Model Inference

KAIST, B.S. in School of computing — Mar. 2018 – Feb. 2023

Publications

Understanding and Exploiting Diagonal Attention Sparsity in Autoregressive Image Generation — Sep. 2026

Daeun Kim, Junwha Hong, Changhun Oh, Yoonsung Kim, Yoonhyeong Lee, Jongse Park

IISWC 2026

Déjà Vu: Efficient Video-Language Query Engine with Learning-based Inter-Frame Computation Reuse — Sep. 2025

Jinwoo Hwang, ***Daeun Kim***, Sangyeop Lee, Yoonsung Kim, Guseul Heo, Hojoon Kim, Yunseok Jeong, Tadiwos Meaza, Eunhyeok Park, Jeongseob Ahn, Jongse Park

VLDB 2025

MixDiT: Accelerating Image Diffusion Transformer Inference with Mixed-Precision MX Quantization — Apr. 2025

Daeun Kim, Jinwoo Hwang, Changhun Oh, Jongse Park

IEEE Computer Architecture Letters (CAL)

CoVA: Exploiting Compressed-Domain Analysis to Accelerate Video Analytics — Jun. 2022

Jinwoo Hwang, Minsu Kim, ***Daeun Kim***, Seungho Nam, Yoonsung Kim, Dohee Kim, Hardik Sharma, Jongse Park

USENIX ATC 2022

Projects

Image generation system with sparse attention: Diagonal-aware Sparse Attention for Autoregressive Image Generation
- github.com/casys-kaist/DiagonalAttn
- Tools used: Pytorch, Triton

Technologies

Languages: Python, C, C++
Tools & Libraries: CUDA, Triton, Pytorch, Diffusers

Experiences

Undergrad Architecture Mentoring (uArch) Workshop mentee, ISCA 2023 Jun. 2023

Undergraduate research intern, KAIST CASys Lab Sep. 2021 - Feb. 2023

- Leveraging sparse CNN in object detection model for accurate and efficient video analytics
- Baseline system development for video analytics

AI engineer intern, Naviworks Co.,Ltd. Mar. 2021 - Aug. 2021

- Training image classification model for military symbols

Teaching Assistant

KAIST CS30300: Operating Systems and Lab Fall 2026

KAIST CS50100: Computer Architecture Spring 2026

KAIST CS230: System Programming Fall 2022

References

Jongse Park jspark@casys.kaist.ac.kr
Professor, KAIST